

Sanchay Narendra Gawande

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Summary

Software Engineer with 2+ years building backend systems and AI infrastructure, strong in **data structures** and **algorithms**, async pipelines, reliability, and performance. Built LLM systems using **RAG**, tool calling, and vector search. Hands on experience building services in **Rust** with **ownership**, **borrow checker**, **async/await**, and **Tokio**.

Technical Skills

- Programming:** Rust · Python · Java · SQL · JavaScript · TypeScript
- Data Structures and Algorithms:** hash maps · trees · graphs · traversal · search algorithms · complexity trade-offs
- Backend:** FastAPI · Node.js · Flask · REST APIs · GraphQL · microservices · async pipelines
- Cloud and Infra:** AWS Lambda · SQS · ECS · API Gateway · S3 · Docker · Kubernetes · GitHub Actions
- Datastores:** PostgreSQL · MongoDB · Redis · Firebase Realtime Database
- AI and LLM:** RAG · Pinecone · OpenAI API · embeddings · LangChain · LangGraph · tool calling · function calling
- ML:** XGBoost · LightGBM · DistilBERT · ONNX Runtime · TensorFlow Serving

Professional Experience

Machine Learning & AI Engineer (Founding Team), Suno Analytics

01/2025 – Present | Remote, USA

- Owned backend design and production delivery of a multi model LLM orchestration service, driving reliability, tail latency, and safe execution across user critical financial workflows.
- Used **Rust** memory safety guarantees and async primitives to improve system reliability and performance, applying ownership driven patterns to reduce shared state bugs under concurrency.
- Built async execution paths using **Rust async/await** patterns to handle concurrent operations safely, keeping service behavior predictable under load and failure conditions.
- Engineered an asynchronous ingestion pipeline using **AWS Lambda** and **SQS** with idempotent processing, retry safety, and backpressure controls, reducing ingestion latency by 40%.
- Implemented deterministic tool calling for transforming Excel workbooks and generating macros, reducing manual intervention and improving output correctness for financial stakeholders.
- Built a stateful retrieval layer using **RAG** with **Pinecone**, metadata routing, and structured tool outputs to keep long running assistants consistent across multi step workflows.
- Improved tail latency by profiling hot paths and reducing bottlenecks, adding request level metrics and structured logs to speed up incident triage and prevent regressions.
- Strengthened production readiness with alerting tied to service level indicators and CI regression detection, reducing recurring failures across releases.

Software Engineer Fellowship, Headstarter

07/2024 – 01/2025 | Remote, USA

- Built multiple production-style applications end-to-end using **Rust**, React, **PostgreSQL**, and **AWS**, shipping features on weekly release cycles.
- Developed backend APIs with clear request contracts, structured error responses, and secure auth patterns to support web user flows.
- Implemented data modeling and query patterns in **SQL** and **PostgreSQL**, focusing on correctness, indexing basics, and predictable performance.
- Created small system-level utilities in **C++** and **Rust** for learning, benchmarking, and performance comparisons against Python services.
- Led peer code reviews and delivered PR level feedback, improving consistency of coding standards and reducing debugging cycles across the cohort.

Software Engineer Intern, University of Massachusetts Boston

01/2024 – 06/2024 | Boston, USA

- Designed a decoupled architecture for a patient guidance platform, separating the React frontend from the Python inference backend for independent scaling.
- Implemented a hybrid search approach combining vector retrieval with metadata rules to improve answer grounding and reduce hallucinations in patient-facing responses.
- Engineered authentication and session state handling to ensure consistent behavior across distributed message sessions and multi-step user interactions.
- Collaborated with stakeholders to translate workflow needs into clear functional requirements and iterative backend deliverables.

Projects

PeerGenius - AI-powered student collaboration platform

- Engineered a context-aware AI response system with 95% accuracy for academic query detection, differentiating solo versus group study modes with secure access using Firebase Authentication and JWT.
- Developed a full-stack architecture with React 19 and Vite frontend, RUST and MongoDB backend, and Firebase Auth integration.

LifeLens - Microservices-based AI daily decision engine

- Built a microservices platform with a Rust backend and FastAPI ML service, integrating 5 or more AI APIs with fallback mechanisms and 95% uptime.
- Engineered **PostgreSQL** with Row Level Security and optimized indexing plus **Redis** caching, improving performance by 30% to 50% with cache invalidation and token optimization, with query times under 200 ms.

Education

University of Massachusetts Boston Boston, MA, USA	Aug 2022 - May 2024
Master of Science in Computer Science	
Sant Gadge Baba Amravati University Maharashtra, India	June 2019 - May 2022
Bachelor of Engineering in Computer Science and Engineering	